THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2008/2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

December 9, 2015 9:30 am – 12:30 pm (3 hours)

Answer **ANY FOUR** questions

All questions carry equal marks

Use of Calculators

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. a) A vessel had hit the Kap Shui Mun bridge and caused minor non-structural damages to the bridge. While the bridge is a crucial route connecting the city and the airport, discuss whether lump sum contract is a suitable arrangement to get the damages repaired. [7 marks]

   b) Propose with justifications another contractual arrangement which can address the problem in (a) more efficiently and effectively. Describe the mechanism of the contractual arrangement being put forward. [10 marks]

   c) The pavement materials of the bridge are due to be completely replaced according to a planned maintenance programme. Should the contract type for this planned maintenance work be the same as that in (a) and/or (b), and if not discuss which is the best contractual arrangement for this type of project? [8 marks]

2. a) An investor is planning to build a 200-unit service apartment. The usable floor area of this service apartment will be 9,000m^2 while the circulation area will take up 20% of the total floor space. The construction cost is estimated to be \$10,000/m^2 and the building will take 24 months to construct. At a rent of \$25,000/unit/month, the service apartment will attract 90% occupancy during its first year of operation before attaining a full occupancy rate subsequently. The initial yield for this type of development would be 6%. With the cost of advertisement being budgeted at \$1,000,000 and a short-term finance at 12% per annum, calculate the maximum site value based on the above parameters and your engineering judgement. [20 marks]

   b) Briefly describe the major drawbacks of residual method and under what circumstances should discount cash flow methods be used for investment appraisal. [5 marks]

3. a) Your company has just been awarded a mega infrastructure project in mainland China. While this is the first and only project in China so far, you are not quite sure which organisational setting should be adopted. Highlight the features of functional and project-based organisational settings and explain their pros and cons. [14 marks]

   b) While matrix organisation is commonly employed by construction companies in general, examine whether this approach is a more appropriate option when applied to this scenario. [7 marks]

   c) Produce an organisation chart based on the organisational setting you have chosen. [4 marks]

4. a) A 10-storey residential building consisting of 20 totally identical units (including 2 units at the ground floor) is going to be built. The project has to be completed within 270 days, and the proposed work sequence and time for the construction of one residential unit are as follows:

Activity	*Duration (week)*
A	1
B	1
C	2
D	1
E	1
F	1
G	1
H	1
I	1

Based on the above data, calculate how long will be the 'turnover' for the activities. [3 marks]

b) Explain what is 'turnover' in the context of construction planning and justify why should a project manager consider the 'turnover' under the circumstances described in (a) above. [5 marks]

c) Prepare a programme (in week) based on the data and findings in (a). [10 marks]

d) Discuss the differences between the programme in (c) and the outline programme and master programme. [7 marks]

5. a) Explain with example(s) what are 'covered price' and 'forming ring' in tendering sense and discuss whether it is ethical for the contractor to submit a tender with a 'covered price' and/or by 'forming ring'? [10 marks]

b) The engineer has discovered that the tender received contains certain qualifications and has been intentionally front loaded. Explain with example(s) what 'qualification' and 'front loading' and how should the engineer deal with these tendering issues? [10 marks]

c) As stated in the tender document, contractors can 'confirm his tender, or withdraw the tender' when tender errors are discovered. Explain the principles of this statement in a lump sum contract setting and discuss whether the same principles can be also applied to a remeasurement contract. [5 marks]

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